

Sharp one-dimensional Lieb–Thirring constants

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Abstract

We resolve affirmatively the remaining cases of the scalar one-dimensional Lieb–Thirring conjecture: for every $\frac{1}{2} < \gamma < \frac{3}{2}$, the optimal constant is the one-bound-state constant. The estimate holds for every nonnegative potential in $L^{\gamma+1/2}(\mathbb{R})$, with all negative eigenvalues included.

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1 Introduction

The Lieb–Thirring inequality bounds the negative eigenvalues of a Schrödinger operator by an integral of its potential. Write $t_+ = \max\{t, 0\}$ and $t_- = \max\{-t, 0\}$. On the line, let $0 \leq W \in L^{\gamma+1/2}(\mathbb{R})$ and let H_{-W} be the self-adjoint operator associated with the quadratic form

$$h_{-W}[v] = \int_{\mathbb{R}} |v'(x)|^2 dx - \int_{\mathbb{R}} W(x)|v(x)|^2 dx, \quad v \in H^1(\mathbb{R}).$$

For $\gamma > 1/2$, this form is closed and bounded below, and its negative spectrum is discrete. We give the needed form and compactness arguments in Section 7. Write

$$\mathrm{Tr}(H_{-W})_-^\gamma = \sum_{\lambda_j(H_{-W}) < 0} |\lambda_j(H_{-W})|^\gamma,$$

counting multiplicities; before the inequality is proved this sum is allowed to be infinite. The optimal constant is

$$L_{\gamma,1} = \sup_{\substack{0 \leq W \in L^{\gamma+1/2}(\mathbb{R}) \\ W \neq 0}} \frac{\mathrm{Tr}(H_{-W})_-^\gamma}{\int_{\mathbb{R}} W^{\gamma+1/2}}.$$

Two natural comparisons are the semiclassical constant

$$L_{\gamma,1}^{\mathrm{cl}} = \frac{1}{2\pi} \int_{\mathbb{R}} (1 - \xi^2)_+^\gamma d\xi = \frac{\Gamma(\gamma+1)}{2\sqrt{\pi} \Gamma(\gamma+3/2)} \quad (1.1)$$

and the optimal constant when only the lowest eigenvalue is retained, denoted by $L_{\gamma,1}^{(1)}$. The one-dimensional Lieb–Thirring conjecture predicts the one-bound-state constant for $1/2 < \gamma < 3/2$ and the semiclassical constant for $\gamma \geq 3/2$.

Theorem 1.1. *For every $1/2 < \gamma < 3/2$ and every nonnegative $W \in L^{\gamma+1/2}(\mathbb{R})$,*

$$\mathrm{Tr}(H_{-W})_-^\gamma \leq 2 \left(\frac{\gamma - 1/2}{\gamma + 1/2} \right)^{\gamma-1/2} L_{\gamma,1}^{\mathrm{cl}} \int_{\mathbb{R}} W^{\gamma+1/2}. \quad (1.2)$$

The constant is optimal and equals $L_{\gamma,1}^{(1)}$. More precisely, if $r = (\gamma - 1/2)^{-1}$, equality is attained by $W(x) = (r+1) \operatorname{sech}^2(rx)$.

The estimate includes rough, unbounded potentials and infinitely many negative eigenvalues. Corollary 7.3 gives the corresponding inequality for signed potentials on their natural dense weighted form domain. The theorem concerns scalar potentials on the line; the matrices used below encode finite orthonormal families. Its sharpness assertion exhibits an equality potential without classifying all optimizers.

1.1 History and relation to prior work

Lieb and Thirring introduced these spectral inequalities in their work on the kinetic energy of fermions and the stability of matter [15, 16]. At exponent one, optimizing over the potential makes the spectral inequality equivalent to a lower bound on the kinetic energy of an orthonormal family in terms of its density. On the line, for a finite orthonormal family $(u_j) \subset H^1(\mathbb{R})$, this density is $\rho = \sum_j |u_j|^2$, and the corresponding kinetic bound controls $\int \rho^3$ by $\sum_j \int |u_j'|^2$. The

one-state variational problem goes back to Keller [12]. The conjecture asks whether allowing arbitrarily many bound states can improve on its constant in the lower exponent range. We verify the one-state equality potential and its normalization directly.

The scalar upper endpoint predates the general Lieb–Thirring inequalities. Inverse-scattering trace identities for the Korteweg–de Vries equation give the sharp value $L_{3/2,1} = 3/16$ [22, 8]; Lieb and Thirring explicitly credit Gardner, Greene, Kruskal and Miura in [16, Section 4(B)]. Aizenman and Lieb’s monotonicity argument extends the semiclassical equality to all larger exponents [1]. At the other endpoint, Weidl first proved finiteness of $L_{1/2,1}$ [21], and Hundertmark, Lieb and Thomas obtained its sharp value $1/2$ [11, Theorem 1]. Laptev and Weidl established the operator-valued upper-endpoint estimate and used dimension lifting to obtain the sharp semiclassical constants for every dimension and $\gamma \geq 3/2$ [13, Theorems 2.1, 2.2 and 3.1].

Quantitative estimates and orthonormal-function methods narrowed the remaining gap. Hundertmark, Laptev and Weidl proved $L_{\gamma,1} \leq 2L_{\gamma,1}^{\text{cl}}$ for $1/2 \leq \gamma < 3/2$ [10, Theorem 4.1]. At exponent one, Dolbeault, Laptev and Loss extended the method of Eden and Foias to matrix-valued potentials [6, 5]. Their all-dimensional semiclassical ratio was $\pi/\sqrt{3}$; subsequent bounds were 1.456 by Frank, Hundertmark, Jex and Nam [7, Theorem 1] and the reported 1.44655 of Corso and Ried [3, Corollary 1.7]. These bounds preceded the sharp exponent-one theorem described below. In the full one-dimensional strict interval, Levitt’s numerical investigation observed convergence to the one-state profile or splitting into increasingly separated copies [14, Section 4.1]. Such computations support the conjectured value but do not bound the contribution of every orthonormal family. That uniform bound is the obstacle addressed by our action inequality.

Read and Schulz proved the case $\gamma = 1$ in [19, Theorem 1], obtaining the sharp constant $4/(3\sqrt{3}\pi)$. Their proof extends Benguria and Loss’s scalar cumulative-mass argument [2, Section 2] to orthonormal families through a matrix correction to the kinetic energy. Their operator-valued extension also gives the semiclassical ratio $2/\sqrt{3}$ in every dimension at exponent one [19, Theorem 2 and Section 2]. Our proof retains the individual eigenvalue scales in a finite-interval action inequality. Its main analytic step is a rank-one comparison for an energy-dependent regularized matrix field. A continuation argument chooses a lower triangular coefficient matrix and a path joining two prescribed matrix endpoints. Together these yield the action inequality for every $1/2 < \gamma < 3/2$. Theorem 1.1 therefore resolves the remaining open-interval cases of the one-dimensional Lieb–Thirring conjecture positively; the case $\gamma = 1$ is already covered by Read–Schulz.

1.2 Proof strategy

Put $\sigma = \gamma - 1/2 \in (0, 1)$. The central result is a variational inequality on a bounded interval, independent of any potential. Fix $I = (x_-, x_+)$. Given N orthonormal real functions $u_i \in H_0^1(I; \mathbb{R})$ and positive numbers k_i , let $u = (u_i)_{i=1}^N$ and $K = \text{diag}(k_i)$. We prove that the action

$$\mathcal{E}_K(a) = \int_I (|a'|^2 + a^\top K^2 a - |a|^{2+2/\sigma}) dx$$

has supremum at least $\sum_i \int_{-k_i}^{k_i} (k_i^2 - s^2)^\sigma ds$ among functions $a = Cu$ with C lower triangular. Section 2 states this precisely in Theorem 2.1 and explains the role of triangularity.

The difficulty is to obtain this lower bound for an arbitrary orthonormal family while retaining its individual scales k_i . The argument replaces a direct optimization over C by a path with prescribed matrix endpoints. Section 3 constructs a regularized map $B \mapsto M_\delta(B)$ from real symmetric $N \times N$ matrices to themselves, and a scalar primitive whose change from K to $-K$

gives the required endpoint integral. Section 4 proves the analytic comparison that controls the nonlinear action term when $M_\delta(B) = aa^\top$. Its integrated tangent map is an average of unitary conjugations; a trace inequality removes this average from the comparison.

Section 5 chooses C and a path from K to $-K$ by continuation modulo row signs. Compactness and the exclusion of zero rows make the continuation possible even when C loses rank. Section 6 combines the path with the primitive: failure of the rank-one constraint contributes a negative square to the action identity. Compactness allows the penalty parameter to tend to zero at fixed regularization. Only then is the regularization removed.

Section 7 returns to the potential. Take the u_i to be Dirichlet eigenfunctions with eigenvalues $-k_i^2$, ordered so that $k_1 \geq \dots \geq k_N$. Lower triangularity lets the eigenfunction equations bound the quadratic part of the action by $\int W|a|^2$. A scalar maximization bounds the whole action by a multiple of $\int W^{1+\sigma}$. Finally, cutoff trial spaces pass from bounded intervals to the line, and finite partial sums exhaust all negative eigenvalues. The sharpness calculation uses the explicit potential in Theorem 1.1.

2 The finite-interval action inequality

Fix

$$0 < \sigma < 1, \quad p = 1 + \sigma, \quad q = 1 + \frac{1}{\sigma}.$$

All matrices below are finite dimensional. Write Sym_N for the real symmetric $N \times N$ matrices, with inner product $\text{tr}(AB)$ and norm $\|A\|_{\text{HS}} = (\text{tr}(A^2))^{1/2}$. The notation $A \leq B$ means that $B - A$ is positive semidefinite. Vectors are columns, and $|a|$ is their Euclidean norm.

Let $I = (x_-, x_+)$ be a bounded interval. For a positive diagonal matrix $K = \text{diag}(k_1, \dots, k_N)$ and $a \in H_0^1(I; \mathbb{R}^N)$, define

$$\mathcal{E}_K(a) = \int_I (|a'|^2 + a^\top K^2 a - |a|^{2q}) dx. \quad (2.1)$$

In one dimension $H_0^1(I)$ embeds into the continuous functions, so every term in this integral is finite.

Theorem 2.1 (Action inequality). *Let $u = (u_1, \dots, u_N)^\top \in H_0^1(I; \mathbb{R}^N)$ satisfy $\int_I uu^\top dx = I_N$, and let $k_i > 0$. Then*

$$\sup_{C \text{ lower triangular}} \mathcal{E}_K(Cu) \geq \sum_{i=1}^N \int_{-k_i}^{k_i} (k_i^2 - s^2)^\sigma ds. \quad (2.2)$$

The supremum is over all real lower triangular $N \times N$ matrices.

The restriction to lower triangular matrices is what makes this inequality useful spectrally. If the u_i are eigenfunctions ordered from the lowest eigenvalue upward, the i th component of Cu stays in the span of the first i eigenfunctions. Section 7 will use this observation to bound the left side from above by an integral of the potential.

Here is the construction behind the lower bound. For each $\delta > 0$, Section 3 builds a locally Lipschitz map $M : \text{Sym}_N \rightarrow \text{Sym}_N$ and a C^1 function $J : \text{Sym}_N \rightarrow \mathbb{R}$ such that

$$dJ(B)[H] = -\text{tr}(M(B)H). \quad (2.3)$$

They have two decisive properties. First, Section 4 proves

$$M(B) = aa^\top \implies a^\top(K^2 - B^2)a \geq |a|^{2q}. \quad (2.4)$$

Second, Section 5 finds, for every $\varepsilon > 0$, a lower triangular C and a path B satisfying

$$\varepsilon B' = M(B) - (Cu)(Cu)^\top, \quad B(x_-) = K, \quad B(x_+) = -K. \quad (2.5)$$

The small parameter ε penalizes failure of the rank-one constraint. For $a = Cu$, differentiating $J(B) + a^\top Ba$ along this path produces the negative term $-\varepsilon^{-1} \|M(B) - aa^\top\|_{\text{HS}}^2$. Uniform bounds on the path and on C then give

$$\sup_C \mathcal{E}_K(Cu) \geq J(-K) - J(K).$$

Finally, the boundary difference tends to the right side of (2.2) as $\delta \downarrow 0$. We keep δ fixed until after the limit $\varepsilon \downarrow 0$.

3 A regularized matrix field

We construct the field and primitive used in Section 2. Throughout this section $K = \text{diag}(k_1, \dots, k_N) > 0$ and $\delta > 0$ are fixed. The scalar case explains the choice of field. For $N = 1$, write $K = (k)$. If $-k \leq b \leq k$, putting $m = (k^2 - b^2)^\sigma$ gives

$$m(k^2 - b^2) = m^{1+1/\sigma}.$$

Thus this field gives equality in the scalar version of the rank-one comparison, and a primitive with derivative $-m$ has the required change $\int_{-k}^k (k^2 - b^2)^\sigma db$ from k to $-k$. We seek matrix versions of the comparison and endpoint identity, with a regularization removed at the end.

The identity

$$k^2 - 2sb + s^2 = (s - b)^2 + k^2 - b^2$$

suggests integrating a scalar function of the quadratic on the left: its affine dependence on b will let us construct a trace primitive even when the corresponding matrices K and B do not commute. Set $\beta = \sigma - \frac{1}{2} \in (-\frac{1}{2}, \frac{1}{2})$ and define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(0) = 0, \quad f'(y) = (\max\{y, 0\} + \delta)^{\beta-1}, \quad c = \frac{\sigma}{\int_{\mathbb{R}} (1 + s^2)^{\beta-1} ds}. \quad (3.1)$$

The exponent and normalization are chosen so that, for $z \geq 0$, scaling the integration variable gives

$$c \int_{\mathbb{R}} f'(s^2 + z) ds = \sigma(z + \delta)^{\beta-1/2} = \sigma(z + \delta)^{\sigma-1}.$$

Integration in z therefore produces the regularized power $(z + \delta)^\sigma - \delta^\sigma$. On the scalar interval $[-k, k]$, we will recover this expression with $z = k^2 - b^2$. Keeping f' constant for negative arguments extends the construction beyond that interval, as needed for the continuation argument. The resulting f is C^1 , strictly increasing, and concave on $\mathbb{R}$. The integral defining c is finite because $\beta < 1/2$; the definition also includes $\beta = 0$ without a limiting convention.

For $B \in \text{Sym}_N$, put

$$\begin{aligned} X_s(B) &= K^2 - 2sB + s^2 I_N, \\ M_R(B) &= c \int_{-R}^R (f(X_s(B)) - f(s^2) I_N) ds, \\ M(B) &= \lim_{R \rightarrow \infty} M_R(B). \end{aligned} \tag{3.2}$$

Here and below scalar functions of symmetric or Hermitian matrices are defined by the spectral theorem. The symmetric cutoff in (3.2) is part of the definition.

3.1 Convergence and a primitive

We first record an elementary estimate that keeps the regularity argument in the Hilbert–Schmidt norm.

Lemma 3.1. *If a real function h is L -Lipschitz on an interval containing the spectra of two Hermitian matrices A and B , then $\|h(A) - h(B)\|_{\text{HS}} \leq L\|A - B\|_{\text{HS}}$.*

Proof. Choose orthonormal eigenbases v_i and w_j of A and B , with respective eigenvalues λ_i and ν_j . In these two bases the entries of $h(A) - h(B)$ and $A - B$ are, respectively,

$$(h(\lambda_i) - h(\nu_j))\langle v_i, w_j \rangle, \quad (\lambda_i - \nu_j)\langle v_i, w_j \rangle.$$

Square, sum over i, j , and apply the scalar Lipschitz bound. $\square$

Lemma 3.2. *The limit in (3.2) is locally uniform. The resulting map M is locally Lipschitz. Moreover,*

$$J(B) = - \int_0^1 \text{tr}(BM(tB)) dt \tag{3.3}$$

is C^1 and satisfies (2.3).

Proof. Fix a bounded set of B 's. For sufficiently large $|s|$, the spectrum of $H_s(B) = K^2 - 2sB$ lies in $[-A|s|, A|s|]$, with A independent of B . On this interval set

$$h_s(t) = f(s^2 + t) - f(s^2) - f'(s^2)t.$$

All arguments of f are positive. Since $f''(y) = (\beta - 1)(y + \delta)^{\beta-2}$ for $y > 0$, the mean value theorem gives

$$\text{Lip}(h_s) = O(|s|^{2\beta-3}), \quad \sup_{|t| \leq A|s|} |h_s(t)| = O(|s|^{2\beta-2}).$$

It follows that

$$f(X_s(B)) - f(s^2)I_N = -2sf'(s^2)B + R_s(B), \tag{3.4}$$

where $R_s(B) = f'(s^2)K^2 + h_s(H_s(B))$ has norm $O(|s|^{2\beta-2})$. By Lemma 3.1, its Lipschitz constant as a function of B has the same bound: the factor $2|s|$ from H_s multiplies $O(|s|^{2\beta-3})$. The first term on the right of (3.4) is odd in s and cancels at every symmetric cutoff; the remainder is integrable. On bounded s -intervals, f is Lipschitz and the same lemma applies directly. This proves both assertions about M .

For completeness we justify the primitive without differentiating M . Let F be any C^2 primitive of f . The trace differentiation formula is

$$d \operatorname{tr} F(A)[H] = \operatorname{tr}(f(A)H). \quad (3.5)$$

For polynomials it follows by cyclicity of trace. Approximation of F and F' on a compact spectral interval by a polynomial and its derivative proves the general formula. For $s \neq 0$, the map $B \mapsto f(X_s(B)) - f(s^2)I_N$ is therefore a gradient: a potential is

$$-\frac{1}{2s} \operatorname{tr} F(X_s(B)) - f(s^2) \operatorname{tr} B.$$

At $s = 0$ the field is constant and is again a gradient. Consequently each M_R has zero integral around every piecewise C^1 closed curve, by integration first along the curve and then in s . Its negative has the primitive $J_R(B) = -\int_0^1 \operatorname{tr}(BM_R(tB)) dt$. Local uniform convergence gives $J_R \rightarrow J$ and $M_R \rightarrow M$ on compact sets. Passing to the limit in the line-segment identity for J_R yields

$$J(B+H) - J(B) = -\int_0^1 \operatorname{tr}(M(B+tH)H) dt.$$

Continuity of M proves that J is C^1 with the stated derivative. $\square$

3.2 Scalar comparison and boundary values

Define

$$\mu(z) = c \int_{\mathbb{R}} (f(s^2 + z) - f(s^2)) ds, \quad z \in \mathbb{R}. \quad (3.6)$$

This integral is absolutely convergent: on bounded z -sets its integrand has tail $O(|s|^{2\beta-2})$.

Lemma 3.3. *The function μ is locally Lipschitz and strictly increasing, with $\mu(0) = 0$. For $z \geq 0$,*

$$\mu(z) = (z + \delta)^\sigma - \delta^\sigma, \quad \mu'(z) = \sigma(z + \delta)^{\sigma-1}. \quad (3.7)$$

For every fixed $b, z \in \mathbb{R}$,

$$c \lim_{R \rightarrow \infty} \int_{-R}^R (f((s-b)^2 + z) - f(s^2)) ds = \mu(z). \quad (3.8)$$

Proof. The tail estimate above, also for differences in z , proves local Lipschitz continuity. Strict increase follows from strict increase of f at every s . Differentiation under the integral for $z \geq 0$ gives

$$\mu'(z) = c \int_{\mathbb{R}} (s^2 + z + \delta)^{\beta-1} ds = \sigma(z + \delta)^{\beta-1/2} = \sigma(z + \delta)^{\sigma-1}.$$

Integration from zero proves (3.7).

For the shift, write $g(s) = f(s^2 + z)$. It is even, and $g'(s) = 2sf'(s^2 + z) = O(|s|^{2\beta-1}) \rightarrow 0$ as $|s| \rightarrow \infty$. Suppose first that $b \geq 0$. The change in the integral of g under the shift is

$$\int_R^{R+b} g(t) dt - \int_{R-b}^R g(t) dt.$$

Its absolute value is at most $b^2 \sup_{|t-R| \leq b} |g'(t)|$, which tends to zero. Evenness handles $b < 0$ as well. Subtracting $f(s^2)$ now proves (3.8). $\square$

Lemma 3.4 (Directional comparison). *For every $B \in \text{Sym}_N$ and every real unit vector e ,*

$$e^\top M(B)e \leq \mu(e^\top K^2 e - (e^\top B e)^2). \quad (3.9)$$

If a standard basis vector e_i is an eigenvector of B with eigenvalue b_i , then

$$M(B)e_i = \mu(k_i^2 - b_i^2)e_i. \quad (3.10)$$

For the primitive in (3.3),

$$J(-K) - J(K) = \sum_{i=1}^N \int_{-k_i}^{k_i} \mu(k_i^2 - s^2) ds. \quad (3.11)$$

Proof. Scalar concavity and the spectral decomposition of $X_s(B)$ give

$$e^\top f(X_s(B))e \leq f(e^\top X_s(B)e).$$

Set $b = e^\top B e$ and $z = e^\top K^2 e - b^2$. The scalar argument on the right is $(s - b)^2 + z$. Integrate at a symmetric cutoff and use Lemma 3.3 to obtain (3.9). This uses scalar concavity only; no operator concavity on the whole real line is asserted.

For (3.10), e_i is also an eigenvector of $X_s(B)$, with eigenvalue $(s - b_i)^2 + k_i^2 - b_i^2$. Functional calculus and (3.8) give the identity. Finally, restrict $dJ = -\text{tr}(M dB)$ to diagonal matrices and integrate along the diagonal segment from K to $-K$. Each coordinate contributes

$$-\int_{k_i}^{-k_i} \mu(k_i^2 - s^2) ds = \int_{-k_i}^{k_i} \mu(k_i^2 - s^2) ds,$$

as claimed. $\square$

In particular $M(K) = M(-K) = 0$. We shall also use the following symmetry. If S is a diagonal matrix with entries ± 1 , then $SKS = K$ and

$$M(SBS) = SM(B)S. \quad (3.12)$$

This follows already at each cutoff. The matrix K stays fixed throughout; we do not require equivariance under conjugations that move K .

4 The rank-one comparison

The field M was chosen so that its rank-one values control the nonlinear term in the action. We now prove that property. The parameters K, σ and δ remain fixed as in Section 3.

Proposition 4.1. *For every $B \in \text{Sym}_N$ and every $a \in \mathbb{R}^N$,*

$$M(B) = aa^\top \implies a^\top (K^2 - B^2)a \geq |a|^{2q}. \quad (4.1)$$

The proof compares $X_s(B)$ with a matrix that is a function of B alone. Its tangent map, integrated in s , is an average of unitary conjugations. Trace monotonicity then removes this average from the needed inequality.

4.1 A tangent inequality on positive matrices

Divided differences describe the derivative of matrix functional calculus, as in the work of Daleckii and Krein [4]; see [9, Theorem 2.3.1] for a modern finite-dimensional formulation. We prove the required derivative formula and tangent inequality directly from the resolvent representation below.

Lemma 4.2. *For $y \geq 0$ the function f of (3.1) has the representation*

$$f(y) = c_\beta \int_0^\infty v^\beta \left(\frac{1}{\delta + v} - \frac{1}{y + \delta + v} \right) dv, \quad c_\beta^{-1} = \int_0^\infty \frac{t^\beta}{(1+t)^2} dt. \quad (4.2)$$

If $X, Y \geq 0$ are Hermitian matrices, then

$$f(X) \leq f(Y) + T_Y(X - Y), \quad (4.3)$$

where, in an eigenbasis of Y with eigenvalues y_i ,

$$(T_Y Z)_{ij} = f[y_i, y_j] Z_{ij}, \quad f[x, y] = \int_0^1 f'(\tau x + (1 - \tau)y) d\tau. \quad (4.4)$$

For $x \neq y$, this is $(f(x) - f(y))/(x - y)$; for $x = y$ it is $f'(x)$.

Proof. The integrand in (4.2) is $O(v^\beta)$ near zero and $O(v^{\beta-2})$ at infinity. The range $-1 < \beta < 1$ suffices for convergence. Differentiation gives

$$c_\beta \int_0^\infty \frac{v^\beta}{(y + \delta + v)^2} dv = (y + \delta)^{\beta-1} = f'(y),$$

and both sides of (4.2) vanish at zero.

Put $A = Y + (\delta + v)I$ and $Z = X - Y$. Since A and $A + Z$ are positive definite, the resolvent identity gives

$$-(A + Z)^{-1} = -A^{-1} + A^{-1}ZA^{-1} - A^{-1}Z(A + Z)^{-1}ZA^{-1} \leq -A^{-1} + A^{-1}ZA^{-1}.$$

Integrating proves (4.3). The coefficient multiplying Z_{ij} in its linear term is

$$c_\beta \int_0^\infty \frac{v^\beta}{(y_i + \delta + v)(y_j + \delta + v)} dv = f[y_i, y_j].$$

This also covers repeated eigenvalues. All these integrals converge absolutely, since $\delta > 0$. □

4.2 The integrated tangent map

Suppose that $M(B) = aa^\top$ and $m = |a|^2 > 0$. By Lemma 3.4, for every unit e ,

$$0 \leq e^\top M(B)e \leq \mu(e^\top K^2 e - (e^\top B e)^2).$$

Strict increase of μ implies $e^\top K^2 e \geq (e^\top B e)^2$. Consequently

$$e^\top X_s(B)e = (s - e^\top B e)^2 + e^\top K^2 e - (e^\top B e)^2 \geq 0 \quad (4.5)$$

for every real s : the matrices $X_s(B)$ are positive semidefinite.

To apply the tangent inequality, we compare $X_s(B)$ with $(sI - B)^2 + dI$, which is diagonal in an eigenbasis of B . Choose $d > 0$ so that its integrated scalar value is $\mu(d) = m$:

$$D = K^2 - B^2, \quad d = (m + \delta^\sigma)^{1/\sigma} - \delta, \quad Y_s = (sI - B)^2 + dI. \quad (4.6)$$

This choice matches the nonzero eigenvalue of $M(B)$. When we pair the integrated tangent inequality with $M(B)$, the identity $M(B)^2 = mM(B)$ will cancel the contribution mI against the left side. We have $X_s(B) = Y_s + (D - dI)$; we do not assert that $D \geq 0$.

Apply Lemma 4.2 to $X_s(B) \geq 0$ and $Y_s > 0$, and denote the tangent map at Y_s by T_s . Subtract $f(s^2)I$ and integrate symmetrically. In an eigenbasis of B , the shift identity (3.8) gives $c \int_{\mathbb{R}} (f(Y_s) - f(s^2)I) ds = mI$, with the integral understood at symmetric cutoffs. Thus

$$M(B) \leq mI + \mu'(d)\mathcal{P}(D - dI), \quad \mathcal{P} = \frac{c}{\mu'(d)} \int_{\mathbb{R}} T_s ds. \quad (4.7)$$

The integral of the tangent maps converges absolutely: for large $|s|$, their matrix coefficients are $O(|s|^{2\beta-2})$.

Lemma 4.3. *In an eigenbasis of B with eigenvalues z_i , the map $\mathcal{P}$ is*

$$(\mathcal{P}Z)_{ij} = P_{ij}Z_{ij}, \quad P_{ij} = \int_0^1 \left(1 + \frac{\tau(1-\tau)(z_i - z_j)^2}{d + \delta} \right)^{\sigma-1} d\tau. \quad (4.8)$$

There is an even probability measure ν on $\mathbb{R}$ such that

$$\mathcal{P}(Z) = \int_{\mathbb{R}} e^{itB} Z e^{-itB} d\nu(t). \quad (4.9)$$

In particular $\mathcal{P}(I) = I$, and $\mathcal{P}$ is self-adjoint for the trace pairing on Hermitian matrices.

Proof. The eigenvalues of Y_s are $y_i = (s - z_i)^2 + d$. Formula (4.4) yields

$$f[y_i, y_j] = \int_0^1 ((s - \tau z_i - (1 - \tau)z_j)^2 + d + \delta + \tau(1 - \tau)(z_i - z_j)^2)^{\beta-1} d\tau.$$

These scalar integrands are nonnegative. Tonelli's theorem and a shift in s evaluate their integral by the same calculation as $\mu'(d)$ in Lemma 3.3, giving (4.8).

For $h \geq 0$ and $x \in \mathbb{R}$, the gamma integral gives

$$(1 + hx^2)^{\sigma-1} = \frac{1}{\Gamma(1 - \sigma)} \int_0^\infty v^{-\sigma} e^{-v} e^{-vhx^2} dv. \quad (4.10)$$

The density $v^{-\sigma} e^{-v} / \Gamma(1 - \sigma)$ has total mass one. For fixed v, h , e^{-vhx^2} is the characteristic function of a centered Gaussian with variance $2vh$; when $h = 0$, use the point mass at zero. Choose τ uniformly in $[0, 1]$ and put $h = \tau(1 - \tau)/(d + \delta)$. The resulting mixture of these symmetric Gaussian measures is an even probability measure ν , independent of i, j . Its characteristic function at $z_i - z_j$ is precisely P_{ij} , which proves (4.9). The diagonal coefficients are one. Finally, changing t to $-t$ in (4.9) and using cyclicity of trace proves self-adjointness. $\square$

The integrated tangent inequality (4.7) controls $\mathcal{P}(D)$, while the desired conclusion involves D itself. The next step compares them in precisely the trace pairing with $M(B)$.

4.3 Trace monotonicity and contraction

Lemma 4.4. *With B, D and $\mathcal{P}$ as above,*

$$\mathrm{tr}(M(B)D) \geq \mathrm{tr}(M(B)\mathcal{P}(D)). \quad (4.11)$$

Proof. We use an elementary instance of the generalized Klein trace inequality [18, Proposition 3, p. 289]: for any increasing real function g and Hermitian matrices A_1, A_2 ,

$$\mathrm{tr}((g(A_1) - g(A_2))(A_1 - A_2)) \geq 0. \quad (4.12)$$

Indeed, in eigenbases of the two matrices the trace equals

$$\sum_{i,j} (g(\lambda_i) - g(\nu_j))(\lambda_i - \nu_j) |\langle v_i, w_j \rangle|^2 \geq 0.$$

Apply this to $g = f$, $A_1 = UX_s(B)U^*$ and $A_2 = X_s(B)$, with $U = e^{itB}$. Since U commutes with $(sI - B)^2$, expansion of the trace gives

$$0 \leq \mathrm{tr}(f(X_s(B))(2D - UDU^* - U^*DU)). \quad (4.13)$$

The coefficient $2D - UDU^* - U^*DU$ has trace zero. We may therefore subtract $f(s^2)I$ inside the trace, integrate over $[-R, R]$, and pass to the symmetric-cutoff limit defining M . The result is

$$0 \leq \mathrm{tr}(M(B)(2D - UDU^* - U^*DU)).$$

This passage is also uniform in t : the coefficient has norm at most $4\|D\|_{\mathrm{HS}}$, whereas $M_R(B) \rightarrow M(B)$ in matrix norm. Average with respect to the even measure ν from Lemma 4.3. Both conjugation terms average to $\mathcal{P}(D)$, proving the claim. $\square$

Proof of Proposition 4.1. If $a = 0$, both sides of (4.1) vanish. Otherwise the preceding construction applies. Multiply the matrix inequality (4.7) by the positive semidefinite matrix $M = aa^\top$ and take its trace. Since $M^2 = mM$ and $\mathcal{P}(I) = I$, we obtain

$$m \mathrm{tr} M \leq m \mathrm{tr} M + \mu'(d)(\mathrm{tr}(M\mathcal{P}(D)) - d \mathrm{tr} M).$$

Here $\mu'(d) > 0$. Lemma 4.4 therefore gives

$$a^\top Da = \mathrm{tr}(MD) \geq \mathrm{tr}(M\mathcal{P}(D)) \geq d \mathrm{tr} M = dm.$$

For nonnegative x, y and $0 < \sigma < 1$, $(x + y)^\sigma \leq x^\sigma + y^\sigma$. Apply this with $x = d$, $y = \delta$ to get $m = (d + \delta)^\sigma - \delta^\sigma \leq d^\sigma$. Thus $dm \geq m^{1+1/\sigma} = |a|^{2q}$, completing the proof. $\square$

5 A matrix path with prescribed endpoints

The action argument needs a symmetric matrix path from K to $-K$. Its differential equation also contains a lower triangular coefficient matrix C , which we are free to choose. We choose C by matching the terminal endpoint. At the start of a deformation, the matching problem has exactly one solution after identifying each row of C with its negative. A compactness argument shows that this solution count cannot disappear modulo two.

Fix $\delta > 0$ throughout the remaining construction. We use the matrix field $M = M_\delta$ and scalar function $\mu = \mu_\delta$ constructed above. Recall the properties needed here. The field M is locally Lipschitz on Sym_N , the space of real symmetric $N \times N$ matrices. The function μ is locally Lipschitz and strictly increasing, with $\mu(0) = 0$. For $K = \text{diag}(k_1, \dots, k_N)$ and every diagonal sign matrix S ,

$$M(SBS) = SM(B)S. \quad (5.1)$$

For every real unit vector e ,

$$e^\top M(B)e \leq \mu(e^\top K^2 e - (e^\top B e)^2). \quad (5.2)$$

Finally, writing e_i for the i th standard basis vector,

$$B e_i = t e_i \implies M(B) e_i = \mu(k_i^2 - t^2) e_i. \quad (5.3)$$

The equivariance in (5.1) only concerns sign matrices, which commute with the fixed matrix K .

Proposition 5.1 (Connecting path). *Let $I = (x_-, x_+)$ be a bounded interval, let $u \in H_0^1(I; \mathbb{R}^N)$ satisfy*

$$\int_I u(x) u(x)^\top dx = I_N,$$

and let $K = \text{diag}(k_1, \dots, k_N)$ with every $k_i > 0$. Suppose M and μ have the regularity and properties (5.1)–(5.3) just stated. For every $\varepsilon > 0$, there are a real lower triangular matrix C and a path $B \in C^1(\bar{I}; \text{Sym}_N)$ such that, with $a = Cu$,

$$\varepsilon B' = M(B) - aa^\top, \quad B(x_-) = K, \quad B(x_+) = -K. \quad (5.4)$$

These choices satisfy

$$\|B(x)\|_{\text{op}} \leq \|K\|_{\text{op}} \quad (x \in I). \quad (5.5)$$

For fixed M , K , and I , the matrices C can be chosen bounded uniformly for $0 < \varepsilon \leq 1$.

The parity principle is the compact-one-manifold argument underlying mod-two degree; see [17, Section 4]. The smooth regular-value facts used below are given in [17, Sections 2–3]. We prove the needed relative version for continuous families of bundle sections here. A section of a vector bundle assigns to each base point a vector in the fiber above it. A zero is *transverse* if the derivative of the section in any local trivialization maps onto that fiber.

Lemma 5.2 (Continuation modulo two). *Let Q be a smooth d -dimensional manifold without boundary, and let $E \rightarrow Q$ be a smooth real vector bundle of rank d . Suppose s_θ , $0 \leq \theta \leq 1$, is a continuous homotopy of sections and*

$$\{(q, \theta) \in Q \times [0, 1] : s_\theta(q) = 0\}$$

is compact. If s_0 is smooth and has an odd number of zeros, all transverse, then s_1 has a zero.

Proof. Suppose that s_1 has no zero. Choose a relatively compact smooth domain $\Omega \subset Q$ containing the projection of the entire zero set. Such a domain is obtained by taking a regular level of a smooth compactly supported function equal to one near that projection. Fix a smooth fiber metric on E . On the compact set

$$(\partial\Omega \times [0, 1]) \cup (\bar{\Omega} \times \{1\}),$$

the norm of the section has a strictly positive minimum.

Reparametrize the homotopy so that it is identically s_0 on a small initial collar. It can then be uniformly approximated on $\overline{\Omega} \times [0, 1]$ by a smooth section that agrees with s_0 on a smaller initial collar. To construct this approximation, use finitely many bundle charts, approximate the continuous coefficient functions by smooth functions, and combine them with a smooth partition of unity. A cutoff supported inside the original collar splices this approximation with the smooth section s_0 . Choose the approximation error smaller than one quarter of the positive boundary minimum. The approximating section therefore has no zeros near the spatial boundary or the final boundary.

We can also make this smooth section transverse to zero while keeping it unchanged near the initial boundary. Here are the finite-dimensional details. In an initial collar its zeros are already transverse, since the derivative in the Q directions is onto at each zero of s_0 . The remaining possible zeros lie in a compact subset of the interior. Choose finitely many compactly supported local frame sections whose values span every fiber on a neighborhood of this subset, with supports away from the initial boundary and the forbidden boundaries. Adding linear combinations of these sections gives a finite-dimensional family whose derivative in the new parameters is onto wherever transversality is still needed. Restrict the parameters to a small open ball in $\mathbb{R}^m$, where m is the number of frame sections. Smallness preserves the positive gap on the compact complement of the spanning neighborhood and the initial collar, so no new zeros appear there. The universal zero set in the interior is a smooth manifold of dimension $m + 1$. Sard's theorem applied to its projection onto $\mathbb{R}^m$ provides arbitrarily small regular parameters. At a zero, write the universal derivative as $Av + Db$, with v a (q, θ) variation and b a parameter variation. Surjectivity of the projection on the kernel means that $Av = -Db$ is solvable for every b . Where perturbations are needed, D is onto, so this is equivalent to A being onto; elsewhere the section is already transverse. Thus regular parameters make the fixed-parameter section transverse. Smallness in the smooth topology preserves the transversality already present on the overlap with the initial collar, and smallness in norm preserves the boundary gap.

The zero set of the resulting section is a compact smooth one-dimensional manifold. Its boundary consists exactly of the zeros of s_0 on $\Omega \times \{0\}$. A compact one-dimensional manifold has an even number of boundary points, contradicting the assumed odd count. No orientation of Q or E is required. $\square$

Proof of Proposition 5.1. Put $\kappa = \|K\|_{\text{op}}$ and $\ell = x_+ - x_-$, and use the continuous representative of u . Choose a smooth nonnegative function χ on Sym_N , invariant under orthogonal conjugation, compactly supported, and equal to one whenever $\|B\|_{\text{op}} \leq \kappa$. For example, take $\chi(B) = \eta(\text{tr } B^2)$ with η equal to one on $[0, N\kappa^2]$ and zero outside a larger bounded interval. The field $\widehat{M} = \chi M$ is bounded and globally Lipschitz.

The terminal matching problem. Let $\mathcal{L}$ denote the vector space of real lower triangular $N \times N$ matrices. For $C \in \mathcal{L}$ and $0 \leq \theta \leq 1$, solve

$$\varepsilon B' = \theta \widehat{M}(B) - (Cu)(Cu)^\top, \quad B(x_-) = K. \quad (5.6)$$

The bounded globally Lipschitz field and continuous forcing give a unique solution throughout I . The integral equation and Gronwall's inequality give continuous dependence on (C, θ) . Thus

$$\Phi_\theta(C) = B(x_+) + K$$

is a continuous homotopy of maps from $\mathcal{L}$ to Sym_N .

At a zero of Φ_θ , integration of (5.6) and orthonormality of u give

$$CC^\top = 2\varepsilon K + \theta \int_I \widehat{M}(B(x)) dx. \quad (5.7)$$

Consequently,

$$\|C\|_{\text{HS}}^2 \leq 2\varepsilon \operatorname{tr} K + \ell \sup_B |\operatorname{tr} \widehat{M}(B)|. \quad (5.8)$$

For fixed ε , the full zero set in $\mathcal{L} \times [0, 1]$ is therefore compact.

Rows cannot vanish at a matching solution. Let G be the group of diagonal sign matrices. Equation (5.1), invariance of χ , and uniqueness of the flow show that

$$\Phi_\theta(SC) = S\Phi_\theta(C)S \quad (S \in G). \quad (5.9)$$

Suppose that row i of C vanishes, and let S_i change only its sign. Then $S_i C = C$, so uniqueness gives $S_i B S_i = B$ throughout the flow. Hence $B e_i = b_i e_i$. By (5.3),

$$\varepsilon b'_i = \theta \chi(B(x)) \mu(k_i^2 - b_i^2), \quad b_i(x_-) = k_i.$$

Regarding $\theta \chi(B(x))/\varepsilon$ as a known continuous coefficient, local Lipschitz uniqueness for this scalar equation gives $b_i \equiv k_i$. Since $k_i > 0$, the terminal value cannot be $-k_i$. Thus no zero of Φ_θ has a vanishing row.

Let $\mathcal{L}^\times \subset \mathcal{L}$ be the open subset on which every row is nonzero. The compact joint zero set lies inside $\mathcal{L}^\times \times [0, 1]$, and hence a positive distance away from every zero-row stratum. We do not require C to be invertible.

One zero modulo signs persists. The action of G on $\mathcal{L}^\times$ is free: changing the sign of a nonzero row cannot fix C . Its quotient

$$Q = \mathcal{L}^\times / G$$

is therefore a smooth manifold of dimension $d = N(N+1)/2$. Concretely, each row i belongs to $\mathbb{R}^i \setminus \{0\}$ and is identified with its negative. Sign conjugation on symmetric matrices defines the rank- d vector bundle

$$E = (\mathcal{L}^\times \times \operatorname{Sym}_N) / G \longrightarrow Q, \quad (C, Z) \sim (SC, SZS).$$

By (5.9), the maps Φ_θ descend to a continuous homotopy of sections of this bundle. Their joint zero set remains compact.

At $\theta = 0$,

$$\Phi_0(C) = 2K - \varepsilon^{-1} CC^\top.$$

The lower triangular solutions of $CC^\top = 2\varepsilon K$ are exactly

$$C = \operatorname{diag}(\pm\sqrt{2\varepsilon k_1}, \dots, \pm\sqrt{2\varepsilon k_N}).$$

Indeed, the first row determines C_{11} , the other entries of the first column then vanish, and the same argument applies successively to the remaining principal blocks. These solutions form one G -orbit. At any one of them, writing $C = \operatorname{diag}(c_i)$,

$$D\Phi_0(C)[H] = -\varepsilon^{-1}(HC^\top + CH^\top) \quad (H \in \mathcal{L}).$$

Its diagonal entries are $-2c_i H_{ii}/\varepsilon$, and its entries with $i > j$ are $-c_j H_{ij}/\varepsilon$. Since every $c_i \neq 0$, this linear map is an isomorphism onto the symmetric matrices. The initial quotient section has exactly one transverse zero. Lemma 5.2 now gives a zero of Φ_1 .

The endpoints remove the cutoff. Take a path obtained from this zero. For a fixed real unit vector e , put $h(x) = e^\top B(x)e$. Whenever $|h(x)| > \kappa$, (5.2) implies

$$\varepsilon h'(x) \leq \chi(B(x))\mu(e^\top K^2 e - h(x)^2) - |e^\top Cu(x)|^2 \leq 0.$$

Here $e^\top K^2 e \leq \kappa^2 < h(x)^2$; the last inequality uses $\chi \geq 0$ and remains true where $\chi = 0$. Both endpoint values of h lie in $[-\kappa, \kappa]$. A component of $\{h > \kappa\}$ cannot start from the initial side: on that component h is nonincreasing, whereas its left endpoint has value κ . Similarly, a component of $\{h < -\kappa\}$ cannot return to the final endpoint: its right endpoint would have value $-\kappa$, again contradicting nonincrease. Thus $|h(x)| \leq \kappa$ throughout I .

Since e was arbitrary, this proves (5.5). Therefore $\chi(B(x)) = 1$ everywhere, and the path solves the uncut equation (5.4). Finally, its integrated equation gives, for $0 < \varepsilon \leq 1$,

$$\|C\|_{\text{HS}}^2 \leq 2 \operatorname{tr} K + \ell \sup_{\|B\|_{\text{op}} \leq \kappa} |\operatorname{tr} M(B)|. \quad (5.10)$$

This is the asserted bound on C . □

For the regularized field M_δ , the bound (5.10) may depend on δ . With δ fixed, it implies that all pairs $(B(x), Cu(x))$ produced for $0 < \varepsilon \leq 1$ lie in one compact subset of $\operatorname{Sym}_N \times \mathbb{R}^N$. This is the compactness needed in the action limit; it requires no bound on B' uniform in ε .

6 From the matrix path to the action

We now prove Theorem 2.1. The matrix comparison controls points where $M(B) = aa^\top$ exactly. The differential equation supplies a penalty for the remaining points, and compactness makes that penalty effective without requiring convergence of the chosen coefficients C .

Proof of Theorem 2.1. Fix $\delta > 0$, and use the field M and primitive J from Section 3. For each $0 < \varepsilon \leq 1$, the connecting-path result of Section 5 provides a lower triangular matrix C_ε and a path B_ε satisfying (2.5). Write $a_\varepsilon = C_\varepsilon u$ and temporarily suppress the subscripts. The path B is C^1 , and $a \in H_0^1(I; \mathbb{R}^N)$ is absolutely continuous. The chain and product rules give, almost everywhere,

$$\begin{aligned} (J(B) + a^\top Ba)' &= 2(a')^\top Ba - \operatorname{tr}((M(B) - aa^\top)B') \\ &= 2(a')^\top Ba - \varepsilon^{-1} \|M(B) - aa^\top\|_{\text{HS}}^2. \end{aligned} \quad (6.1)$$

Since a vanishes at the endpoints, integration and $2(a')^\top Ba \leq |a'|^2 + a^\top B^2 a$ yield

$$J(-K) - J(K) \leq \mathcal{E}_K(a) + \int_I F_\varepsilon(B(x), a(x)) dx, \quad (6.2)$$

where

$$F_\varepsilon(B, a) = |a|^{2q} - a^\top (K^2 - B^2)a - \varepsilon^{-1} \|M(B) - aa^\top\|_{\text{HS}}^2.$$

The bounds from Section 5 place every pair $(B_\varepsilon(x), a_\varepsilon(x))$, for $x \in \bar{I}$ and $0 < \varepsilon \leq 1$, in a single compact set $\mathcal{K}$. This uses the continuous representative of u , the bound on C_ε , and fixed δ . We claim that

$$\eta_\varepsilon := \max_{(B,a) \in \mathcal{K}} \max\{F_\varepsilon(B,a), 0\} \longrightarrow 0. \quad (6.3)$$

Otherwise there would be $\eta > 0$, a sequence $\varepsilon_n \downarrow 0$, and a convergent sequence $(B_n, a_n) \in \mathcal{K}$ with $F_{\varepsilon_n}(B_n, a_n) \geq \eta$. The first two terms of F_ε are uniformly bounded on $\mathcal{K}$, so $M(B_n) - a_n a_n^\top \rightarrow 0$. At the limit (B_*, a_*) we have $M(B_*) = a_* a_*^\top$ and

$$|a_*|^{2q} - a_*^\top (K^2 - B_*^2) a_* \geq \eta,$$

contrary to Proposition 4.1. This proves (6.3).

Equation (6.2) now implies

$$J(-K) - J(K) \leq \sup_{C \text{ lower triangular}} \mathcal{E}_K(Cu) + |I|\eta_\varepsilon.$$

Let $\varepsilon \downarrow 0$ at this fixed δ . By Lemma 3.4,

$$\sup_C \mathcal{E}_K(Cu) \geq \sum_i \int_{-k_i}^{k_i} ((k_i^2 - s^2 + \delta)^\sigma - \delta^\sigma) ds.$$

Finally let $\delta \downarrow 0$. On each integration interval the integrand converges to $(k_i^2 - s^2)^\sigma$ and lies between zero and this same function, by subadditivity of the power σ . Dominated convergence gives (2.2). No convergence of C_ε , and no bound uniform in δ , has been used. $\square$

7 From the action inequality to spectral moments

We apply Theorem 2.1 first to ordered Dirichlet eigenfunctions. We then pass to the line using finite-dimensional trial spaces. This last step treats all negative eigenvalues, without assuming in advance that their moment is finite.

Throughout this section, $0 < \sigma < 1$, $p = 1 + \sigma$, $q = 1 + 1/\sigma$, and $\gamma = \sigma + 1/2$. Write

$$A_\sigma = \frac{\sigma^\sigma}{(1 + \sigma)^{1+\sigma}}, \quad \ell_\sigma = \frac{A_\sigma}{B(1/2, 1 + \sigma)}, \quad (7.1)$$

where B denotes the beta function.

7.1 Quadratic forms and negative spectrum

For $0 \leq W \in L^p(\mathbb{R})$, define

$$h_{-W}[v] = \int_{\mathbb{R}} |v'|^2 - \int_{\mathbb{R}} W|v|^2, \quad v \in H^1(\mathbb{R}).$$

We record the facts needed to define its operator and to exhaust its negative eigenvalues. We use the association of a densely defined closed semibounded form with its self-adjoint operator, and the spectral description of the operator and its form domain, in [20, Theorem 2.13, Theorems 3.7–3.8 and (3.50)–(3.51)]. All theorem numbers here refer to the 2009 first edition.

Lemma 7.1. *Let $p > 1$ and $0 \leq W \in L^p(\mathbb{R})$. The form h_{-W} is closed and bounded below on $H^1(\mathbb{R})$. For its associated self-adjoint operator H_{-W} , every spectral projection onto $(-\infty, -\alpha]$, with $\alpha > 0$, has finite rank. Thus its negative spectrum consists of eigenvalues of finite multiplicity, with possible accumulation only at zero. On a bounded interval I , the corresponding Dirichlet form on $H_0^1(I)$ is closed and bounded below and has compact resolvent.*

Proof. Put $p' = p/(p-1)$. The one-dimensional Sobolev inequality and interpolation give

$$\|v\|_{2p'}^2 \leq \|v\|_\infty^{2/p} \|v\|_2^{2-2/p} \leq 2^{1/p} \|v'\|_2^{1/p} \|v\|_2^{2-1/p}.$$

Consequently Hölder's and Young's inequalities imply that, for every $\eta > 0$, there is $b_{\eta, W} < \infty$ such that

$$\int_{\mathbb{R}} W|v|^2 \leq \eta \|v'\|_2^2 + b_{\eta, W} \|v\|_2^2. \quad (7.2)$$

The KLMN Theorem [20, Theorem 6.24] now gives closure and semiboundedness, with form domain exactly $H^1(\mathbb{R})$.

Multiplication by $\sqrt{W}$ is compact from $H^1(\mathbb{R})$ to $L^2(\mathbb{R})$. Indeed, choose nonnegative bounded, compactly supported W_n converging to W in L^p . Multiplication by $\sqrt{W_n}$ is compact by local Rellich compactness, and

$$\|(\sqrt{W} - \sqrt{W_n})v\|_2^2 \leq \|W - W_n\|_p \|v\|_{2p'}^2 \leq C_p \|W - W_n\|_p \|v\|_{H^1}^2.$$

Thus these multiplication operators converge in operator norm.

If the spectral subspace of H_{-W} for $(-\infty, -\alpha]$ were infinite-dimensional, it would contain an orthonormal sequence v_n . Semiboundedness places this entire spectral subspace in the form domain. The lower bound obtained from (7.2) with $\eta = 1/2$, together with $h_{-W}[v_n] \leq -\alpha$, bounds this sequence in H^1 . After passing to a subsequence, it converges weakly to zero in H^1 , because it is orthonormal in L^2 . Compactness then gives $\sqrt{W}v_n \rightarrow 0$ in L^2 , contradicting

$$-\alpha \geq h_{-W}[v_n] \geq -\|\sqrt{W}v_n\|_2^2.$$

This proves the negative-spectrum assertion. On a bounded interval, the same form estimate follows by zero extension. The form norm is equivalent to the H_0^1 norm, whose embedding into $L^2(I)$ is compact; hence the Dirichlet operator has compact resolvent. $\square$

7.2 The Dirichlet estimate

The order of the eigenfunctions is what makes the triangular coefficient restriction in Theorem 2.1 useful.

Proposition 7.2. *Let I be a bounded interval and $0 \leq W \in L^p(I)$. If $-k_1^2, \dots, -k_N^2$ are the first N negative Dirichlet eigenvalues of $-d^2/dx^2 - W$, ordered so that $k_1 \geq \dots \geq k_N > 0$, then*

$$\sum_{j=1}^N k_j^{2\gamma} \leq \ell_\sigma \int_I W^p. \quad (7.3)$$

Proof. Choose real orthonormal eigenfunctions $u_1, \dots, u_N$; these exist because the Dirichlet form is real. Put $u = (u_1, \dots, u_N)^\top$ and $K = \text{diag}(k_1, \dots, k_N)$. For every real lower triangular matrix

C , write $a = Cu$. Its i th component lies in the span of $u_1, \dots, u_i$, and the eigenfunction equations in form sense give

$$\int_I (|a'_i|^2 - W|a_i|^2 + k_i^2|a_i|^2) = \sum_{j \leq i} C_{ij}^2 (k_i^2 - k_j^2) \leq 0.$$

Summing over i and putting $m = |a|^2$, we obtain

$$\mathcal{E}_K(a) \leq \int_I (Wm - m^q) \leq A_\sigma \int_I W^{1+\sigma}.$$

The last inequality follows by maximizing $Wm - m^q$ over $m \geq 0$; the maximizer is $(W/q)^\sigma$.

Theorem 2.1 therefore yields

$$\sum_{j=1}^N \int_{-k_j}^{k_j} (k_j^2 - s^2)^\sigma ds \leq A_\sigma \int_I W^{1+\sigma}.$$

Substituting $s = k_j t$ in each integral gives $B(1/2, 1 + \sigma) k_j^{2\sigma+1}$, proving (7.3). $\square$

The beta–gamma identity identifies the constant without any limiting argument in σ :

$$\begin{aligned} \ell_\sigma &= \left(\frac{\sigma}{1+\sigma} \right)^\sigma \frac{\Gamma(\sigma + 3/2)}{\sqrt{\pi} \Gamma(\sigma + 2)} \\ &= 2 \left(\frac{\sigma}{1+\sigma} \right)^\sigma L_{\gamma,1}^{\text{cl}}. \end{aligned} \tag{7.4}$$

7.3 Exhaustion of the line

Let $0 \leq W \in L^p(\mathbb{R})$. If there are no negative eigenvalues, the trace inequality is immediate. Otherwise fix any finite number N of them, counted with multiplicity and ordered increasingly:

$$\lambda_1 \leq \dots \leq \lambda_N < 0.$$

Let $\phi_1, \dots, \phi_N \in H^1(\mathbb{R})$ be corresponding orthonormal eigenfunctions. Choose smooth cutoffs χ_R equal to one on $[-R, R]$, supported in $I_R = (-2R, 2R)$, and satisfying $0 \leq \chi_R \leq 1$ and $\|\chi'_R\|_\infty \leq c/R$. Then

$$v_{j,R} = \chi_R \phi_j \longrightarrow \phi_j \quad \text{in } H^1(\mathbb{R}).$$

Hölder's inequality and the embedding $H^1 \subset L^{2p'}$ show that the potential form is continuous under this convergence.

For clarity, let G_R and Q_R be the Gram and form matrices of the N functions $v_{j,R}$. Then

$$G_R \longrightarrow I_N, \quad Q_R \longrightarrow \Lambda := \text{diag}(\lambda_1, \dots, \lambda_N)$$

in matrix norm. In particular the functions are linearly independent for large R . If $g_R = \|G_R - I_N\|_{\text{op}} < 1$ and $e_R = \|Q_R - \Lambda\|_{\text{op}}$, their Rayleigh quotients differ uniformly from the limiting coefficient-space quotients by at most

$$\eta_R = \frac{e_R + \|\Lambda\|_{\text{op}} g_R}{1 - g_R} \longrightarrow 0.$$

Applying the form-domain min–max principle [20, Theorem 4.10 and the following form-domain extension] to the span of $v_{1,R}, \dots, v_{j,R}$ gives, for every $j \leq N$,

$$\lambda_j(I_R) \leq \lambda_j + \eta_R,$$

where $\lambda_j(I_R)$ is the j th Dirichlet eigenvalue on I_R . All these levels are negative for sufficiently large R . Proposition 7.2 now implies

$$\sum_{j=1}^N (-\lambda_j - \eta_R)^\gamma \leq \sum_{j=1}^N |\lambda_j(I_R)|^\gamma \leq \ell_\sigma \int_{I_R} W^p \leq \ell_\sigma \int_{\mathbb{R}} W^p.$$

Letting $R \rightarrow \infty$ proves the estimate for every finite partial sum. Taking their supremum gives

$$\mathrm{Tr}(H_{-W})_-^\gamma \leq \ell_\sigma \int_{\mathbb{R}} W^p. \quad (7.5)$$

Thus the full moment is finite even when there are infinitely many negative eigenvalues. No approximation of W by smoother potentials is needed.

7.4 Signed potentials

To include unbounded positive parts, we specify the form domain in the signed-potential statement.

Corollary 7.3. *Let V be real measurable with $V_- \in L^p(\mathbb{R})$, and suppose*

$$\mathcal{D}_V = \left\{ v \in H^1(\mathbb{R}) : \int_{\mathbb{R}} V_+ |v|^2 < \infty \right\}$$

is dense in $L^2(\mathbb{R})$. This holds in particular if $V_+ \in L^1_{\mathrm{loc}}(\mathbb{R})$. The form

$$h_V[v] = \int_{\mathbb{R}} |v'|^2 + \int_{\mathbb{R}} V_+ |v|^2 - \int_{\mathbb{R}} V_- |v|^2, \quad v \in \mathcal{D}_V,$$

is closed and bounded below. Its associated operator H_V has discrete negative spectrum and satisfies

$$\mathrm{Tr}(H_V)_-^\gamma \leq \ell_\sigma \int_{\mathbb{R}} V_-^p.$$

Proof. The positive form $\int |v'|^2 + \int V_+ |v|^2$ is closed on $\mathcal{D}_V$, by completeness of H^1 and of the weighted L^2 space. Estimate (7.2), with $W = V_-$, proves closure and semiboundedness after subtraction of the negative part. If $V_+ \in L^1_{\mathrm{loc}}$, then $C_c^\infty(\mathbb{R}) \subset \mathcal{D}_V$, proving the asserted density. The compactness argument in Lemma 7.1 still applies. On a negative spectral subspace bounded away from zero, the same relative form bound controls the H^1 norm of unit vectors, since the positive potential term can be discarded. Compactness of multiplication by $\sqrt{V_-}$ then excludes an infinite orthonormal sequence in that subspace: the integral against V_- would tend to zero along a subsequence, whereas its negative form energy stays bounded away from zero. Thus every such spectral projection has finite rank.

For $v \in \mathcal{D}_V \subset H^1(\mathbb{R})$, one has $h_V[v] \geq h_{-V_-}[v]$. The min-max principle, with this inclusion of form domains, bounds each negative eigenvalue of H_V below by the corresponding negative eigenvalue of H_{-V_-} ; in particular, if H_V has j negative eigenvalues, so does H_{-V_-} . Summation and (7.5) prove the result. $\square$

7.5 Sharpness

Put $r = 1/\sigma > 1$ and consider

$$W(x) = (r+1) \operatorname{sech}^2(rx), \quad \psi(x) = \operatorname{sech}^{1/r}(rx).$$

Direct differentiation gives

$$\frac{\psi'}{\psi} = -\tanh(rx), \quad \frac{\psi''}{\psi} = 1 - (r+1)\operatorname{sech}^2(rx).$$

The function ψ and its first two derivatives decay exponentially. Thus $\psi \in H^2(\mathbb{R})$, the operator domain for this bounded W , and $H_{-W}\psi = -\psi$. Substitution $t = rx$, followed by $z = \tanh t$, gives

$$\int_{\mathbb{R}} W^p = \frac{(r+1)^p}{r} B(1/2, p), \quad \left(\int_{\mathbb{R}} W^p \right)^{-1} = \ell_{\sigma}.$$

The eigenvalue -1 gives $\operatorname{Tr}(H_{-W})_{-}^{\gamma} \geq 1$; the upper bound (7.5) gives the reverse inequality. This completes the proof of Theorem 1.1. Hence this potential attains equality, and the constant in (7.4) is optimal. This establishes $L_{\gamma,1} = L_{\gamma,1}^{(1)}$ throughout $1/2 < \gamma < 3/2$.

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